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AIRCRAFT CONTENT DISTRIBUTION NETWORK WITH DISTRIBUTED STORAGE SYSTEM

Abstract

A content distribution network (CDN) network with a distributed storage system, and associated devices and methods, are disclosed herein. In some implementations, a content distribution network system for an aircraft includes onboard servers and in-flight-entertainment (IFE) devices on the aircraft. Each onboard server can include central applications, a first CDN node, and a central database operably. Each IFE device can include IFE applications, a second CDN node, and an IFE database. The central databases and the IFE databases can form a distributed storage system that is configured to store multimedia content and/or packets of multimedia content in one or more of the central databases and/or the IFE databases. The central applications can be configured to access the distributed storage system via the first and/or second CDN nodes, and the IFE applications can be configured to access the distributed storage system via the first and/or second CDN nodes.

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Background/Summary

TECHNICAL FIELD

[0001] This patent document is directed generally to systems, methods, and apparatuses for managing a content distribution network with a distributed storage system onboard an aircraft.

BACKGROUND

[0002] Commercial aircraft today often include in-flight entertainment (IFE) devices that can interact with passengers to provide various forms of entertainment (e.g., movies, music, TV shows, etc.), flight-tracking programs, and other services. The multimedia content and software applications providing those services are often stored on ground-based servers and onboard servers physically on the aircraft. However, accessing data stored in ground-based servers requires connectivity and sufficient bandwidth, which is hardly guaranteed on commercial aircraft. Moreover, onboard servers can be costly and limited in storage space, making it difficult to store all of the data locally on the aircraft. Therefore, there is a need for an improved storage system that can simultaneously accommodate larger amounts of data and provide such data in a reliable manner.

SUMMARY

[0003] An aircraft-based content distribution network that includes a distributed storage arrangement are disclosed.

[0004] In one example aspect, a content distribution system is disclosed. The system includes one or more onboard servers on the aircraft, each of the onboard servers comprising one or more central applications; a first content distribution network (CDN) node operably coupled to the one or more central applications; and a central database operably coupled to the first CDN node; and one or more in-flight-entertainment (IFE) devices on the aircraft. Each of the IFE devices comprises one or more IFE applications; a second CDN node operably coupled to the one or more IFE applications; and an IFE database operably coupled to the second CDN node. The central databases and the IFE databases form a distributed storage system. The distributed storage system is configured to store multimedia content and/or packets of multimedia content in one or more of the central databases and/or the IFE databases. The one or more central applications are configured to access the distributed storage system via the first and/or second CDN nodes. The one or more IFE applications are configured to access the distributed storage system via the first and/or second CDN nodes.

[0005] In another aspect, a distributed storage system onboard an aircraft is disclosed. The system includes one or more central databases included in one or more onboard servers on the aircraft, wherein the central databases are operably coupled to a first content distribution network (CDN) node included in the one or more onboard servers; and one or more in-flight entertainment (IFE) databases included in one or more IFE devices on the aircraft, wherein the IFE databases are operably coupled to a second CDN node included in the one or more IFE devices, wherein the one or more central databases and the one or more IFE databases are configured to store multimedia content and/or packets of multimedia content, and wherein the one or more central databases and the one or more IFE databases are configured to provide requested content to applications included in at least one of the one or more onboard servers or the one or more IFE devices via the first and/or second CDN nodes.

[0006] In yet another aspect, a method of operating a content distribution network system (is disclosed. The method includes receiving a request for content from an application onboard the aircraft; accessing a distributed storage system on the aircraft, wherein the distributed storage system comprises: one or more central databases included in one or more onboard servers on the aircraft; and one or more in-flight entertainment (IFE) databases included in one or more IFE

devices on the aircraft; retrieving the requested content from at least one of the central databases and/or the IFE databases of the distributed storage system; and transferring the retrieved content to the application.

[0007] In yet another aspect, a hardware platform comprising at least one processor for implementing the above-described method is disclosed.

[0008] In yet another aspect, a computer-readable medium (CRM) is disclosed. The CRM stores code, which, upon execution by one or more processors, causes the one or more processor to implement the above-described method.

[0009] These, and other, aspects are further described throughout the present document.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0010] Features, aspects, and advantages of the presently disclosed technology may be better understood with regard to the following drawings.

[0011] FIG. 1 shows an example of in-flight entertainment (IFE) devices installed in an airplane based on some implementations of the disclosed technology.

[0012] FIG. 2 shows an example block diagram of a computing device based on some implementations of the disclosed technology.

[0013] FIG. 3 is a schematic diagram of a content distribution network with a distributed storage system based on some implementations of the disclosed technology.

[0014] FIG. 4 is a flowchart illustrating a method for operating a content distribution network system for an aircraft based on some implementations of the disclosed technology.

[0015] A person skilled in the relevant art will understand that the features shown in the drawings are for purposes of illustrations, and variations, including different and/or additional features and arrangements thereof, are possible.

DETAILED DESCRIPTION

[0016] Various implementations will be discussed in detail with reference to the figures below. In the description, the technology is described with respect to aircraft such as commercial planes, but the implementations of the disclosed technology can be applicable to other vehicles such as jets, buses, trains, ships, and other types of passenger vehicles.

1. Introduction

[0017] Various implementations of the disclosed technology provide techniques for expanding the effective amount of total storage space onboard an aircraft for storing multimedia content and other forms of data. Multimedia content (e.g., movies, music, TV shows, etc.), flight data, cabin data, software applications for processing such content and data, etc. are conventionally stored on ground-based servers and onboard servers physically on the aircraft. However, accessing data stored in ground-based servers from an aircraft in-flight requires reliable connectivity and sufficient bandwidth, which is hardly guaranteed. Relying on onboard servers can be similarly problematic, as airplanes have limited space, cooling, and power for maintaining such onboard servers. Moreover, onboard servers can be costly to install and maintain, making it difficult to store all of the data locally on the aircraft. Furthermore, the amount of data demanded and the content access speed expected by passengers continue to increase. Therefore, there is a growing need for an improved storage system on aircraft that can simultaneously accommodate larger amounts of data and provide such data in a reliable manner.

[0018] In some implementations of the present technology, a content distribution network system for an aircraft includes one or more onboard servers and one or more in-flight-entertainment (IFE) devices on the aircraft. Each of the onboard servers can include one or more central applications, a first content distribution network (CDN) node operably coupled to the one or more central

applications, and a central database operably coupled to the first CDN node. Each of the IFE devices can include one or more IFE applications, a second CDN node operably coupled to the one or more IFE applications, and an IFE database operably coupled to the second CDN node. The central databases and the IFE databases can form a distributed storage system that is configured to store multimedia content and/or packets of multimedia content in one or more of the central databases and/or the IFE databases. The central applications can be configured to access the distributed storage system via the first and/or second CDN nodes, and the one or more IFE applications can be configured to access the distributed storage system via the first and/or second CDN nodes.

[0019] In some implementations, a distributed storage system onboard an aircraft includes one or more central databases included in one or more onboard servers on the aircraft, and one or more in-flight entertainment (IFE) databases included in one or more IFE devices on the aircraft. The central databases can be operably coupled to a first content distribution network (CDN) node included in the one or more onboard servers, and the IFE databases can be operably coupled to a second CDN node included in the one or more IFE devices. The one or more central databases and the one or more IFE databases can be configured to store multimedia content and/or packets of multimedia content, and provide requested content to applications included in at least one of the one or more onboard servers or the one or more IFE devices via the first and/or second CDN nodes.

2. Examples of Aircraft-Based CDNs

[0020] FIG. 1 shows an example of a content distribution network (CDN) **110** for an airplane **102** based on some implementations of the disclosed technology. The CDN **110** includes a plurality of in-flight entertainment (IFE) devices **108**, which can provide various entertainment and connectivity services, including video and audio streaming and Internet communications, to passengers on board. For example, the IFE devices **108** can include such as monitors, handsets, ports for transferring audio/power/data, etc. The IFE devices **108** can also communicate or otherwise interact with personal electronic devices (PEDs) carried by passengers. PEDs may refer to any electronic computing device that includes one or more processors or circuitries for implementing the functions related to data storage, video and audio streaming, wired communications, wireless communications, etc. Examples of the PEDs include cellular phones, smart phones, tablet computers, laptop computers, and other portable computing devices. In implementations of the disclosed technology, PEDs may have the capability to execute application software programs (“apps”) to perform various functions and/or communicate with the IFE devices **108**.

[0021] In the illustrated embodiment, the passenger seats are individually labeled Seat**11** to Seat **66**. In some implementations, the IFE devices **108** are provided at each passenger seat, such as located at each of the seatbacks of the passenger seats, and/or on cabin walls and/or deployable from an armrest for seats located at a bulkhead (i.e., in the first row of a section). The IFE devices **108** can include displays providing interfaces to each passenger through which each passenger enters their selections on the entertainment option, e.g., the particular selections, emergency requests, etc. Upon receiving the selection from the passengers, based on the selections from the passengers, the IFE devices **108** can display entertainment content and travel information.

[0022] The CDN **110** can also include one or more onboard servers **122** communicably coupled to the IFE devices **108** and/or the PEDs via one or more wireless access points **120**. The onboard servers **122** can perform various operations including transferring and managing data as discussed with reference to FIGS. 3 and 4. The communications between the servers **122**, the IFE devices **108**, and the PEDs are realized by wired and/or wireless connections. In some implementations, the communication among the servers **122**, the IFE devices **108**, and the PEDs are achieved through the antenna **124** to and from a ground-based server **114** and/or ground-based cell towers by, for example, a provision of network plugs at the seat for the IFE devices **108** to a wired onboard local area network. In some other implementations, the communications among the servers **122**, the IFE

devices **108**, and the PEDs are achieved through the antenna **126** to and from satellites **130**, **132**, **134** in an orbit (e.g., via a cellular network utilizing one or more onboard base station(s), Wi-Fi utilizing the wireless access point **120**, and/or Bluetooth).

[0023] The servers **122**, the IFE devices **108**, and the PEDs form a local network part of the CDN **110** onboard the airplane **102** through an onboard router (not shown). The servers **122** are also communicably coupled with the ground server **114** through the antenna **124** for receiving and transmitting information from/to the ground server **114**. The ground server **114** can be located at various locations, such as a computer center at an arbitrary location on the ground, etc. The ground server **114** may be in communication with a database **116**, provide information from the database **116** to the server **122**, and store information received from the server **122** in the database **116**. Although FIG. **1** shows that the database **116** is provided separately from the ground server **114**, the database **116** can be provided as a part of the ground server **114**.

3. Examples of Hardware Platforms

[0024] FIG. **2** shows an example block diagram of a computing device **200** (e.g., an onboard server, a ground server, or a portable server) based on some implementations of the disclosed technology. The computing device **200** includes at least one processor **201**, a memory **203**, a transceiver **210**, a control module **220**, a database **230**, and an input/output (I/O) interface **240**. In other embodiments, additional, fewer, and/or different elements may be used to configure the computing device **200**. The memory **203** may store instructions and applications to be executed by the processor **201**. The memory **203** is an electronic holding place or storage for information or instructions so that the information or instructions can be accessed by the processor **201**. The memory **203** can include, but is not limited to, any type of random access memory (RAM), any type of read-only memory (ROM), any type of flash memory, such as magnetic storage devices (e.g., hard disk, floppy disk, magnetic strips, etc.), optical disks (e.g., compact disc (CD), digital versatile discs (DVD), etc.), smart cards, flash memory devices, etc. The instructions upon execution by the processor **201** configure the computing device **200** to perform the operations (e.g., the operations as shown in and described with reference to FIGS. **3** and **4**), which will be described in this patent document. The instructions executed by the processor **201** may be carried out by a special purpose computer, logic circuits, or hardware circuits. The processor **201** may be implemented in hardware, firmware, software, or any combination thereof. The term “execution” is, for example, the process of running an application or the carrying out of the operation called for by an instruction. The instructions may be written using one or more programming language, scripting language, assembly language, etc. By executing the instruction, the processor **201** can perform the operations called for by that instruction.

[0025] The processor **201** operably couples with the memory **203**, the transceiver **210**, the control module **220**, the database **230**, and the I/O interface **240**, to receive, send, and process information and to control the operations of the computing device **200**. The processor **201** may retrieve a set of instructions from a permanent memory device, such as a ROM device, and copy the instructions in an executable form to a temporary memory device that is generally some form of RAM. In some implementations, the computing device **200** can include a plurality of processors that use the same or a different processing technology. The transceiver **210** may include a transmitter and a receiver. In some embodiments, the device **200** comprises a transmitter and a receiver that are separate from one another but functionally form a transceiver. The transceiver **210** transmits or sends information or data to another device (e.g., the IFE devices **108**, a reader device, etc.) and receives information or data transmitted or sent by another device (e.g., another server, a PAD, etc.).

[0026] The control module **220** of the computing device **200** is configured to perform operations to assist the computing device **200**. In some implementations, the control module **220** can be configured as a part of the processor **201**. When the computing device **200** communicates with the IFE devices **108** in FIG. **1**, the control module **220** can be included in the airplane **102**. In some implementations, the control module **220** can operate machine learning/artificial intelligence (AI)

applications that perform various types of data analysis to automate analytical model building. Using algorithms that iteratively learn from data, machine learning applications can enable computers to learn without being explicitly programmed. The machine learning/AI module may be configured to use data learning algorithms to build models to interpret various data received from the various devices or components to detect, classify, and/or predict future outcomes. Such data learning algorithms may be associated with rule learning, artificial neural networks, inductive logic programming, and/or clustering. In some implementations, the control module **220** may assist the computing device **200** to perceive their environment and take actions that maximize the effectiveness of the operations performed by the computing device **200**.

[0027] The I/O interfaces **240** enable data to be provided to the computing device **200** as input and enable the computing device **200** to provide data as output. In some embodiments, the I/O interfaces **240** may enable user input to be obtained and received by the computing device **200** (e.g., via a touch-screen display, buttons, or switches) and may enable the computing device **200** to display information. In some embodiments, devices, including touch screen displays, buttons, controllers, audio speakers, or others, are connected to the computing device **200** via I/O interfaces **240**.

[0028] In some implementations, the computing device **200** comprises a computer-readable medium (e.g., the memory **203**) including processor instructions that, when executed by one or more processors (e.g., the processor **201**), cause the one or more processors to perform data storage, transfer, and management processes as described in further detail below with reference to FIGS. **3** and **4**. In some embodiments, the computing device **200** may be used as a hardware platform for the CDN node **324** or **334**.

4. Examples of Virtualized CDN Operation

[0029] FIG. **3** is a schematic diagram of a content distribution network (CDN) system **300** with a distributed storage system **340** based on some implementations of the disclosed technology. The CDN system **300** can be an example of the CDN **110** illustrated in and described above with reference to FIG. **1**. In the illustrated embodiment, the CDN system **300** includes one or more onboard servers **320** physically located on the aircraft **102**, one or more IFE devices **108** also physically located on the aircraft **102**, and one or more ground servers **114** located off board the aircraft **102**.

[0030] Each ground server **114** can include one or more CDN nodes (individually labeled **314a**, **314b**, **314c**, collectively referred to as “the CDN nodes **314**”). The CDN nodes **314** can be operably coupled to or otherwise associated with ground-based databases (not shown) and/or can connect to the Internet.

[0031] Each onboard server **320** can include one or more central applications **322**, a CDN node **324**, a CDN ingest manager **326**, and a central database **328**. The central applications **322** can include a wide range of programs that provide in-flight entertainment (e.g., movies, music, TV shows), support flight operations (e.g., providing navigation and weather data), assist flight crew members (e.g., allowing communication with passengers, checking various statuses), and more. The CDN node **324** can include a device or point through which data is transferred and/or processed. In some implementations, the CDN node **324** is a virtual storage instance that can interact with other nodes in a networked environment, as discussed in further detail below. The CDN ingest manager **326** can include a logic, processor, or other component with a decision-making algorithm, and a communication module. In some implementations, the CDN ingest manager **326** includes an artificial intelligence and/or machine learning model. In some implementations, at least one of the onboard servers **320** includes the CDN ingest manager **326**, which can be operably coupled to the CDN node **324** and the central database **328** of one or more of the onboard servers **320**. The central database **328** can include one or more hard drives, solid-state drives (SSDs), USB flash drives, memory cards, network-attached storage (NAS) devices, etc.

[0032] Each IFE device **108** can include one or more IFE applications **332**, a CDN node **334**, and

an IFE database **338**. The IFE applications **332** can include programs specifically designed or configured to run on the IFE devices **108**. For example, the IFE applications **332** can include software programs for playing multimedia content, flight tracking, shopping, Internet connectivity, etc. The IFE applications **332** can also manage user interfaces that allow passengers to select which of the central applications **322** and/or the IFE applications **332** to run on their dedicated IFE device **108**. The CDN node **334** can include a device or point through which data is transferred and/or processed. In some implementations, the CDN node **324** is a virtual storage instance that can interact with other nodes in a networked environment, as discussed in further detail below. The IFE database **338** can include one or more hard drives, solid-state drives (SSDs), USB flash drives, memory cards, network-attached storage (NAS) devices, etc.

[0033] When storing data onboard the airplane **102**, the storage space available on the central databases **328** of the onboard servers **320** and the storage space available on the IFE databases **338** of the IFE devices **108** can be pooled together such that the central databases **328** and the IFE databases **338** form a single virtual disk, referred to as a distributed storage system **340** hereinafter. The distributed storage system **340** can store data by, for example, storing a complete data packet (e.g., an entire movie) in one of the central databases **328** or one of the IFE databases **338**, or storing parts of a data packet (e.g., parts of a movie) in multiple ones of the central databases **328** and/or the IFE databases **338**. The onboard servers **320** and/or the IFE devices **108**, and/or other components on the airplane **102** can include processors configured to perform segmentation and stitching of data packets such that data can be properly stored in pieces and eventually assembled when provided to passengers.

[0034] In operation, the CDN system **300** in accordance with implementations of the present technology can receive requests for content or other forms of data and provide such content accordingly. For example, when a passenger submits a request to watch a certain movie to one of the IFE applications **332** (e.g., via a user interface on a display included in the IFE devices **108**), the request can be passed from the IFE application **332** to the CDN node **334** and/or the CDN node **324**. In some implementations, the CDN node **324** and the CDN node **334** are synchronized such that they can read data from the central database **328** and the IFE database **338**, respectively, and determine whether the requested data is fully available on the distributed storage system **340**, whether stored on a single database or segmented in multiple databases. If the requested movie is found to be fully available on the distributed storage system **340**, the movie can be extracted, copied, or otherwise delivered to the IFE application **332** via the CDN node **324** and/or the CDN node **334**.

[0035] However, if the requested movie is found not to be fully available on the distributed storage system **340**, the CDN node **324** can submit a cache miss request to the CDN ingest manager **326** (e.g., to a communication module included in the CDN ingest manager **326**). In response, the CDN ingest manager **326** can communicate (e.g., via the communication module included in the CDN ingest manager **326**) with the ground servers **114** to receive the requested movie (or missing parts thereof) from one or more of the CDN nodes **314** of the ground servers **114**. In some implementations, the data transfer between the CDN nodes **314** and the CDN ingest manager **326** is done via satellite communication, High Frequency Data Link (HF DL), and/or other forms of wireless communication protocols. After or while receiving the data packets from the CDN nodes **314**, the CDN ingest manager **326** can decide whether and/or how to store the data packets in the distributed storage system **340**. For example, the CDN ingest manager **326** can determine which database would be appropriate based on the data type, data size, and/or other factors, which databases are available to store additional data, whether one more databases should be offloaded (e.g., remove data from the distributed storage system **340**), whether and/or how to segment the data prior to storage, what level of content redundancy to maintain, etc. In some implementations, the CDN ingest manager **326** and the CDN node **324** are synchronized such that they can communicate data transfer progress or other statuses in real-time.

[0036] In another example, one of the central applications **322** can make a request for data to the CDN node **324**, the CDN node **324** can sync with the CDN node **334** to determine whether the requested data is available on the distributed storage system **340**, submit a cache miss request to the CDN ingest manager **326** as needed, and provide the requested data to the central application **322**. The steps taken can be generally similar to the steps described above with respect to the requested received from the IFE application **332**.

[0037] In some implementations, the CDN ingest manager **326** can be configured to anticipate demands for certain types of data, and coordinate with the CDN nodes **314**, **324**, **334** accordingly. For example, the CDN ingest manager **326** can anticipate what multimedia content the passengers will likely request based on passenger data, the flight route, etc. The CDN ingest manager **326** can then coordinate with the CDN nodes **314**, **324**, **334** to ensure that the anticipated content is available on the distributed storage system **340** prior to passengers actually making such requests. In some implementations, the CDN ingest manager **326** can include a machine learning model trained on training sets that provide passenger data, flight route data, etc. as inputs and specific requests for content as expected outputs.

[0038] The CDN system **300** including the distributed storage system **340** in accordance with implementations of the present technology is expected to significantly improve the experiences of passengers interacting with IFE devices **108**, crew members managing flight operations, etc. First, the total available storage space available on the distributed storage system **340** is generally equal to the sum of the storage space available on all of the central databases **328** on the onboard servers **320** and the storage space available on all of the IFE databases **338** on the IFE devices **108**. Therefore, using the distributed storage system **340** offers significantly increased storage space available compared to just using the onboard servers **320** for data storage, as conventional aircraft do. This means that a significantly greater amount of multimedia content (and other content) and/or higher quality content (e.g., movies of higher resolutions, flight tracking programs with more detail) can be readily available onboard the airplane **102**.

[0039] Having more content locally available means that it is less likely that the airplane **102** will need to retrieve requested or anticipated content from the ground servers **114**, which may be slow, costly, unreliably, and/or unavailable. Less usage of bandwidth for streaming movies (as an example) also means that more bandwidth remains available for other activities such as passenger Internet browsing. Having more content locally available also translates to reduced overall latency for passengers in receiving the requested content.

[0040] Moreover, because the distributed storage system **340** is accessed via the CDN nodes **324**, **334**, the central applications **322** and the IFE applications **332** need not be modified to adapt to the presence of the distributed storage system **340** on the airplane **102**. In other words, the CDN nodes **324**, **334** provide a level of abstraction between the applications **322**, **332** and the distributed storage system **340** such that the applications **322**, **332** can access and benefit from the distributed storage system **340** without even being aware of (e.g., without the need to include compatibility features or add-ons, hardware/software modifications, etc.) the distributed storage system **340**.

[0041] In addition, in implementations in which the CDN ingest manager **326** is able to anticipate requests for content and pre-store the content anticipated to be in demand on the distributed storage system **340** accordingly, the CDN system **300** is expected to further reduce latency for passengers who eventually make requests for such content.

5. Examples of Operating an Aircraft CDN

[0042] FIG. **4** is a flowchart illustrating a method **400** for operating a content distribution network system for an aircraft based on some implementations of the disclosed technology. While the method **400** is described below with reference to the embodiments of FIGS. **1-3**, those skilled in the art will appreciate that the method **400** can be practiced with other embodiments of the present technology. Moreover, the steps described herein can be practiced in an order different from the order in which they are presented. The method **400** may also omit some of the steps described

herein and/or include additional steps.

[0043] The method **400** begins at block **402** by receiving a request for content from an application (e.g., the central applications **322**, the IFE applications **332**) onboard an aircraft (e.g., the aircraft **102**). The application can be on an onboard server of the aircraft or on an IFE device of the aircraft. The requested content can be multimedia content (e.g., movies, music, TV shows, etc.), flight tracking programs, recommended travel itineraries, shopping content, cabin data, etc.

[0044] At block **404**, the method **400** continues by accessing a distributed storage system (e.g., the distributed storage system **304**) on the aircraft. The distributed storage system can include one or more central databases (e.g., the central databases **328**) included in one or more onboard servers (e.g., the onboard servers **320**) on the aircraft, and one or more in-flight entertainment (IFE) databases (e.g., the IFE databases **338**) included in one or more IFE devices (e.g., the IFE devices **108**) on the aircraft.

[0045] At block **406**, the method **400** continues by retrieving the requested content from at least one of the central databases and/or the IFE databases of the distributed storage system. At block **408**, the method **400** continues by transferring the retrieved content to the application. In some implementations, content distribution network (CDN) nodes included in the onboard servers or the IFE devices (e.g., CDN nodes **324**, CDN nodes **334**) perform the steps of blocks **406** and/or **408**.

[0046] In some implementations, the method **400** can further include anticipating requests for content by the application, determining that the distributed storage system does not store the content anticipated to be requested by the application, and receiving the content anticipated to be requested by the application from one or more ground servers (e.g., the ground servers **114**).

[0047] In some implementations, the method **400** can further include receiving content from one or more ground servers, and determining in which of the central databases and/or the IFE databases to store the received content. In some implementations, determining in which of the central databases and/or the IFE databases to store the received content is based on at least one of a type of the received content, a file size of the received content, remaining storage space on the central databases, or remaining storage space on the IFE databases.

[0048] In some implementations, the method **400** can further include determining that the distributed storage system does not store the requested content, and submitting a cache miss request to a processor (e.g., the CDN ingest manager **326**) included in the one or more onboard servers. The processor can be configured to, in response to the cache miss request, retrieve the requested content from one or more ground servers.

[0049] In some implementations, the method **400** can further include determining that the requested content is available on at least one of the central databases and/or the IFE databases of the distributed storage system.

[0050] In accordance with the disclosed technology herein, some preferred embodiments may incorporate technical solutions as indicated in the following examples:

[0051] 1. A content distribution network system for an aircraft (e.g., as described with respect to FIG. **1** and FIG. **3**), the system comprising: one or more onboard servers on the aircraft, each of the onboard servers comprising one or more central applications; a first content distribution network (CDN) node operably coupled to the one or more central applications; and a central database operably coupled to the first CDN node; and one or more in-flight-entertainment (IFE) devices on the aircraft. Each of the IFE devices comprises one or more IFE applications; a second CDN node operably coupled to the one or more IFE applications; and an IFE database operably coupled to the second CDN node. The central databases and the IFE databases form a distributed storage system. The distributed storage system is configured to store multimedia content and/or packets of multimedia content in one or more of the central databases and/or the IFE databases. The one or more central applications are configured to access the distributed storage system via the first and/or second CDN nodes. The one or more IFE applications are configured to access the distributed storage system via the first and/or second CDN nodes.

[0052] 2. The system of any one of the examples herein, wherein at least one of the one or more onboard servers further comprises a CDN ingest manager operably coupled to the first CDN node and the distributed storage system, wherein the CDN ingest manager includes: a communication module configured to receive content from one or more ground servers; and a processor operably coupled to the communication module and configured to determine in which of the central databases and/or the IFE databases the content received by the communication module is to be stored.

[0053] 3. The system of any one of the examples herein, wherein the processor of the CDN ingest manager is further configured to: anticipate requests for content from the one or more central applications and/or the one or more IFE applications; determine that the distributed storage system does not store the content anticipated to be requested by the one or more central applications and/or the one or more IFE applications; and receive the content anticipated to be requested by the one or more central applications and/or the one or more IFE applications from the one or more ground servers.

[0054] 4. The system of any one of the examples herein, wherein the processor of the CDN ingest manager is further configured to: anticipate requests for content from the one or more central applications and/or the one or more IFE applications; and determine that the distributed storage system does store the content anticipated to be requested by the one or more central applications and/or the one or more IFE applications.

[0055] 5. The system of any one of the examples herein, wherein the processor of the CDN ingest manager is further configured to segment the content received by the communication module into a plurality of segments and store the segments in one or more of the central databases and/or the IFE databases.

[0056] 6. The system of any one of the examples herein, wherein the CDN ingest manager includes a machine learning model trained to anticipate requests for content by the one or more central applications and/or the one or more IFE applications based on at least one of passenger data or flight route of the aircraft.

[0057] 7. The system of any one of the examples herein, wherein at least one of the one or more onboard servers further comprises a CDN ingest manager operably coupled to the first CDN node and the distributed storage system, wherein the first CDN node is configured to: receive a request for content from the one or more central applications and/or the one or more IFE applications; determine that the distributed storage system does not store the content requested; and submit a cache miss request to the CDN ingest manager.

[0058] 8. The system of any one of the examples herein, wherein the CDN ingest manager is configured to, in response to the cache miss request, retrieve the content requested from one or more ground servers.

[0059] 9. The system of any one of the examples herein, wherein each of the first and second CDN nodes is configured to: receive a request for content from the one or more central applications and/or the one or more IFE applications; determine that the distributed storage system does store the content requested; and transfer the content requested from at least one of the central databases and/or the IFE databases to the one or more central applications and/or the one or more IFE applications.

[0060] 10. The system of any one of the examples herein, wherein the first and second CDN nodes are operably coupled and synchronized such that each of the first and second CDN nodes can determine content stored in the central databases and in the IFE databases.

[0061] 11. A distributed storage system onboard an aircraft (e.g., FIGS. 1 to 3), the system comprising: one or more central databases included in one or more onboard servers on the aircraft, wherein the central databases are operably coupled to a first content distribution network (CDN) node included in the one or more onboard servers; and one or more in-flight entertainment (IFE) databases included in one or more IFE devices on the aircraft, wherein the IFE databases are

operably coupled to a second CDN node included in the one or more IFE devices, wherein the one or more central databases and the one or more IFE databases are configured to store multimedia content and/or packets of multimedia content, and wherein the one or more central databases and the one or more IFE databases are configured to provide requested content to applications included in at least one of the one or more onboard servers or the one or more IFE devices via the first and/or second CDN nodes.

[0062] 12. The system of any one of the examples herein, wherein the one or more central databases and the one or more IFE databases are configured to receive content to store from ground servers through a processor included in the one or more onboard servers, wherein the processor is configured to determine in which of the central databases and/or the IFE databases the content received is to be stored.

[0063] 13. The system of any one of the examples herein, wherein the first and second CDN nodes are operably coupled and synchronized such that each of the first and second CDN nodes can determine content stored in the central databases and in the IFE databases.

[0064] 14. The system of any one of the examples herein, wherein the first and second CDN nodes provide a level of abstraction such that the applications included in at least one of the one or more onboard servers or the one or more IFE devices do not include compatibility features configured to access content stored in the system.

[0065] 15. A method of operating a content distribution network system (e.g., FIG. 4) for an aircraft, the method comprising: receiving a request for content from an application onboard the aircraft; accessing a distributed storage system on the aircraft, wherein the distributed storage system comprises: one or more central databases included in one or more onboard servers on the aircraft; and one or more in-flight entertainment (IFE) databases included in one or more IFE devices on the aircraft; retrieving the requested content from at least one of the central databases and/or the IFE databases of the distributed storage system; and transferring the retrieved content to the application.

[0066] 16. The method of any one of the examples herein, further comprising: anticipating requests for content by the application; determining that the distributed storage system does not store the content anticipated to be requested by the application; and receiving the content anticipated to be requested by the application from one or more ground servers.

[0067] 17. The method of any one of the examples herein, further comprising: receiving content from one or more ground servers; and determining in which of the central databases and/or the IFE databases to store the received content.

[0068] 18. The method of any one of the examples herein, wherein determining in which of the central databases and/or the IFE databases to store the received content is based on at least one of a type of the received content, a file size of the received content, remaining storage space on the central databases, or remaining storage space on the IFE databases.

[0069] 19. The method of any one of the examples herein, further comprising: determining that the distributed storage system does not store the requested content; and submitting a cache miss request to a processor included in the one or more onboard servers, wherein the processor is configured to, in response to the cache miss request, retrieve the requested content from one or more ground servers.

[0070] 20. The method of any one of the examples herein, further comprising: determining that the requested content is available on at least one of the central databases and/or the IFE databases of the distributed storage system.

6. Concluding Remarks

[0071] It will be appreciated by one of skill in the art that the present document discloses techniques that may be used to solve practical problems related to storage and distribution of multimedia data on an aircraft. In one example aspect, onboard caching is performed of data that, in prior art systems, would have to be retrieved from a storage that is off-site to the airplane.

Furthermore, this data may be cached in a distributed manner. The disclosed arrangement therefore results in saving of valuable communication resources and further reduces computing and other electro-mechanical (e.g., cooling and storage weight and size) resource needed to provide a satisfactory user experience during travel.

[0072] Some of the embodiments described herein are described in the general context of methods or processes, which may be implemented in one embodiment by a computer program product, embodied in a computer-readable medium, including computer-executable instructions, such as program code, executed by computers in networked environments. A computer-readable medium may include removable and non-removable storage devices including, but not limited to, Read Only Memory (ROM), Random Access Memory (RAM), compact discs (CDs), digital versatile discs (DVD), etc. Therefore, the computer-readable media can include a non-transitory storage media. Generally, program modules may include routines, programs, objects, components, data structures, etc. that perform particular tasks or implement particular abstract data types. Computer- or processor-executable instructions, associated data structures, and program modules represent examples of program code for executing steps of the methods disclosed herein. The particular sequence of such executable instructions or associated data structures represents examples of corresponding acts for implementing the functions described in such steps or processes.

[0073] Some of the disclosed embodiments can be implemented as devices or modules using hardware circuits, software, or combinations thereof. For example, a hardware circuit implementation can include discrete analog and/or digital components that are, for example, integrated as part of a printed circuit board. Alternatively, or additionally, the disclosed components or modules can be implemented as an Application Specific Integrated Circuit (ASIC) and/or as a Field Programmable Gate Array (FPGA) device. Some implementations may additionally or alternatively include a digital signal processor (DSP) that is a specialized microprocessor with an architecture optimized for the operational needs of digital signal processing associated with the disclosed functionalities of this application. Similarly, the various components or sub-components within each module may be implemented in software, hardware or firmware. The connectivity between the modules and/or components within the modules may be provided using any one of the connectivity methods and media that is known in the art, including, but not limited to, communications over the Internet, wired, or wireless networks using the appropriate protocols.

[0074] While this document contains many specifics, these should not be construed as limitations on the scope of an invention that is claimed or of what may be claimed, but rather as descriptions of features specific to particular embodiments. Certain features that are described in this document in the context of separate embodiments can also be implemented in combination in a single embodiment. Conversely, various features that are described in the context of a single embodiment can also be implemented in multiple embodiments separately or in any suitable subcombination. Moreover, although features may be described above as acting in certain combinations and even initially claimed as such, one or more features from a claimed combination can in some cases be excised from the combination, and the claimed combination may be directed to a sub-combination or a variation of a sub-combination. Similarly, while operations are depicted in the drawings in a particular order, this should not be understood as requiring that such operations be performed in the particular order shown or in sequential order, or that all illustrated operations be performed, to achieve desirable results.

[0075] Only a few implementations and examples are described, and other implementations, enhancements and variations can be made based on what is described and illustrated in this patent document.

Claims

1. A content distribution network system for an aircraft, the system comprising: one or more onboard servers on the aircraft, each of the onboard servers comprising: one or more central applications; a first content distribution network (CDN) node operably coupled to the one or more central applications; and a central database operably coupled to the first CDN node; and one or more in-flight-entertainment (IFE) devices on the aircraft, each of the IFE devices comprising: one or more IFE applications; a second CDN node operably coupled to the one or more IFE applications; and an IFE database operably coupled to the second CDN node, wherein the central databases and the IFE databases form a distributed storage system, wherein the distributed storage system is configured to store multimedia content and/or packets of multimedia content in one or more of the central databases and/or the IFE databases, wherein the one or more central applications are configured to access the distributed storage system via the first and/or second CDN nodes, and wherein the one or more IFE applications are configured to access the distributed storage system via the first and/or second CDN nodes.
2. The system of claim 1, wherein at least one of the one or more onboard servers further comprises a CDN ingest manager operably coupled to the first CDN node and the distributed storage system, wherein the CDN ingest manager includes: a communication module configured to receive content from one or more ground servers; and a processor operably coupled to the communication module and configured to determine in which of the central databases and/or the IFE databases the content received by the communication module is to be stored.
3. The system of claim 2, wherein the processor of the CDN ingest manager is further configured to: anticipate requests for content from the one or more central applications and/or the one or more IFE applications; determine that the distributed storage system does not store the content anticipated to be requested by the one or more central applications and/or the one or more IFE applications; and receive the content anticipated to be requested by the one or more central applications and/or the one or more IFE applications from the one or more ground servers.
4. The system of claim 2, wherein the processor of the CDN ingest manager is further configured to: anticipate requests for content from the one or more central applications and/or the one or more IFE applications; and determine that the distributed storage system does store the content anticipated to be requested by the one or more central applications and/or the one or more IFE applications.
5. The system of claim 2, wherein the processor of the CDN ingest manager is further configured to segment the content received by the communication module into a plurality of segments and store the segments in one or more of the central databases and/or the IFE databases.
6. The system of claim 2, wherein the CDN ingest manager includes a machine learning model trained to anticipate requests for content by the one or more central applications and/or the one or more IFE applications based on at least one of passenger data or flight route of the aircraft.
7. The system of claim 1, wherein at least one of the one or more onboard servers further comprises a CDN ingest manager operably coupled to the first CDN node and the distributed storage system, wherein the first CDN node is configured to: receive a request for content from the one or more central applications and/or the one or more IFE applications; determine that the distributed storage system does not store the content requested; and submit a cache miss request to the CDN ingest manager.
8. The system of claim 7, wherein the CDN ingest manager is configured to, in response to the cache miss request, retrieve the content requested from one or more ground servers.
9. The system of claim 1, wherein each of the first and second CDN nodes is configured to: receive a request for content from the one or more central applications and/or the one or more IFE applications; determine that the distributed storage system does store the content requested; and transfer the content requested from at least one of the central databases and/or the IFE databases to the one or more central applications and/or the one or more IFE applications.

- 10.** The system of claim 1, wherein the first and second CDN nodes are operably coupled and synchronized such that each of the first and second CDN nodes can determine content stored in the central databases and in the IFE databases.
- 11.** A distributed storage system onboard an aircraft, the system comprising: one or more central databases included in one or more onboard servers on the aircraft, wherein the central databases are operably coupled to a first content distribution network (CDN) node included in the one or more onboard servers; and one or more in-flight entertainment (IFE) databases included in one or more IFE devices on the aircraft, wherein the IFE databases are operably coupled to a second CDN node included in the one or more IFE devices, wherein the one or more central databases and the one or more IFE databases are configured to store multimedia content and/or packets of multimedia content, and wherein the one or more central databases and the one or more IFE databases are configured to provide requested content to applications included in at least one of the one or more onboard servers or the one or more IFE devices via the first and/or second CDN nodes.
- 12.** The system of claim 11, wherein the one or more central databases and the one or more IFE databases are configured to receive content to store from ground servers through a processor included in the one or more onboard servers, wherein the processor is configured to determine in which of the central databases and/or the IFE databases the content received is to be stored.
- 13.** The system of claim 11, wherein the first and second CDN nodes are operably coupled and synchronized such that each of the first and second CDN nodes can determine content stored in the central databases and in the IFE databases.
- 14.** The system of claim 11, wherein the first and second CDN nodes provide a level of abstraction such that the applications included in at least one of the one or more onboard servers or the one or more IFE devices do not include compatibility features configured to access content stored in the system.
- 15.** A method of operating a content distribution network system for an aircraft, the method comprising: receiving a request for content from an application onboard the aircraft; accessing a distributed storage system on the aircraft, wherein the distributed storage system comprises: one or more central databases included in one or more onboard servers on the aircraft; and one or more in-flight entertainment (IFE) databases included in one or more IFE devices on the aircraft; retrieving the requested content from at least one of the central databases and/or the IFE databases of the distributed storage system; and transferring the retrieved content to the application.
- 16.** The method of claim 15, further comprising: anticipating requests for content by the application; determining that the distributed storage system does not store the content anticipated to be requested by the application; and receiving the content anticipated to be requested by the application from one or more ground servers.
- 17.** The method of claim 15, further comprising: receiving content from one or more ground servers; and determining in which of the central databases and/or the IFE databases to store the received content.
- 18.** The method of claim 17, wherein determining in which of the central databases and/or the IFE databases to store the received content is based on at least one of a type of the received content, a file size of the received content, remaining storage space on the central databases, or remaining storage space on the IFE databases.
- 19.** The method of claim 15, further comprising: determining that the distributed storage system does not store the requested content; and submitting a cache miss request to a processor included in the one or more onboard servers, wherein the processor is configured to, in response to the cache miss request, retrieve the requested content from one or more ground servers.
- 20.** The method of claim 15, further comprising: determining that the requested content is available on at least one of the central databases and/or the IFE databases of the distributed storage system.
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